Confidence Intervals & Hypothesis Testing

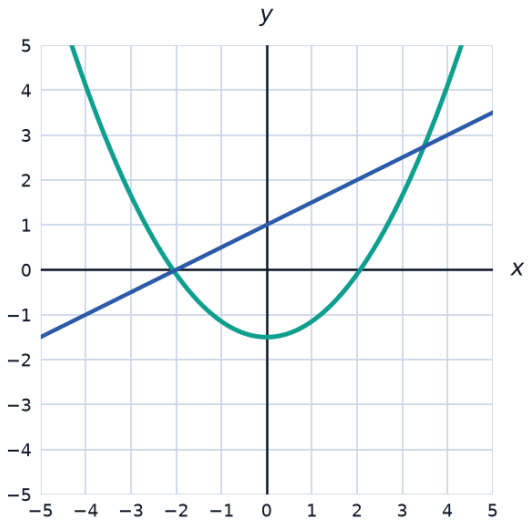
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Constructing a confidence interval

A random sample of 50 students has a mean study time of $\bar{x} = 6.2$ hours per week, with sample standard deviation $s = 1.8$ hours. Construct a 95% confidence interval for the population mean study time, using a critical value of $z^* \approx 1.96$.

2 [Foundational] • Target: Constructing a confidence interval

Interpret the confidence interval from the previous problem in the context of the study time question, using correct statistical language.



3 [Foundational] • Target: Margin of error and sample size

Explain two ways to decrease the margin of error of a confidence interval, and state a tradeoff involved with each.

4 [Foundational] • Target: The logic of hypothesis testing

State the null hypothesis H_0 and alternative hypothesis H_a for a test of whether a coin is biased toward heads (true proportion of heads p), and classify the test as one-tailed or two-tailed.

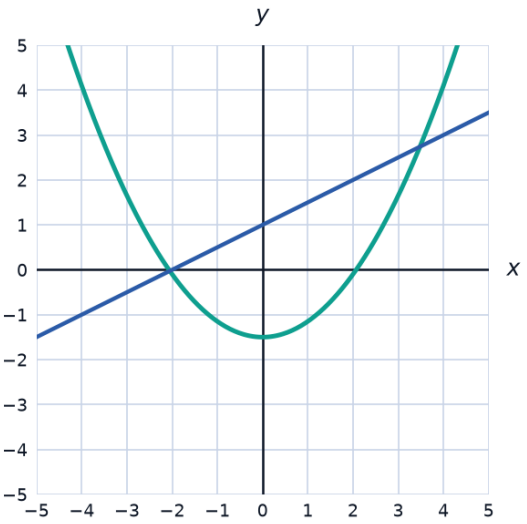
5 [Foundational] • Target: The logic of hypothesis testing

Explain what a p-value represents, in the context of a hypothesis test.

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6 [Foundational] • Target: Four-step hypothesis testing procedure

A manufacturer claims their light bulbs last on average $\mu = 1000$ hours. A sample of 40 bulbs has mean lifetime $\bar{x} = 985$ hours with sample standard deviation $s = 45$ hours. State the hypotheses and compute the test statistic for testing whether the true mean lifetime is less than claimed.



7 [Foundational] • Target: Four-step hypothesis testing procedure

Using the test statistic $z \approx -2.11$ from the previous problem and a p-value of approximately 0.017, state a conclusion at the $\alpha = 0.05$ significance level.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Type I and Type II errors, and power

In the light bulb test above, describe what a Type I error would mean in context, and what a Type II error would mean in context.

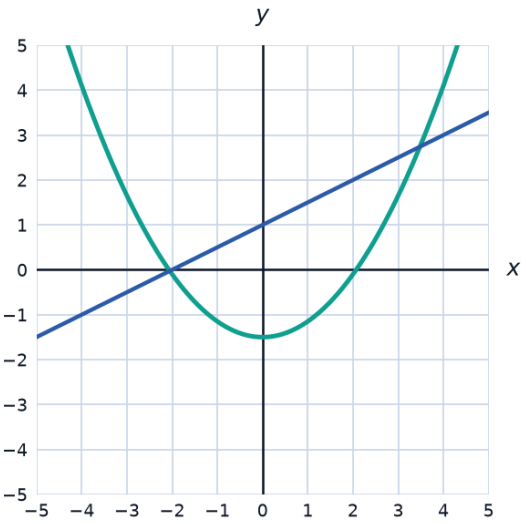
9 [Intermediate] • Target: Type I and Type II errors, and power

Explain what the power of a test measures, and state one way to increase it.

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10 **[Intermediate]** • **Target: Confidence intervals for a proportion**

A poll of 400 randomly selected voters finds that 220 support a candidate. Construct a 95% confidence interval for the true population proportion, using $z^* \approx 1.96$.



11 **[Intermediate]** • **Target: Determining sample size for a target margin of error**

A researcher wants a margin of error of at most 0.03 for a 95% confidence interval for a population proportion, using a conservative planning value of $p = 0.5$ (which maximizes required sample size). Find the minimum sample size needed, using $n = \left(\frac{z^*}{m}\right)^2 p(1-p)$ with $z^* = 1.96$.

12 **[Intermediate]** • **Target: One-sample t-test for a mean**

A nutritionist claims a new granola bar has a mean of 200 calories. A sample of 15 bars has mean $\bar{x} = 206$ calories with sample standard deviation $s = 9$ calories. Since the population standard deviation is unknown and n is small, a t-test is used. Compute the test statistic $t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}$.

13 **[Intermediate]** • **Target: One-sample t-test for a mean**

Explain what it means for a hypothesis test result to be "statistically significant," and why statistical significance does not always imply practical importance.

14 **[Intermediate]** • **Target: One-sample t-test for a mean**

A researcher wants to test whether a new fitness program changes the average resting heart rate of participants from the population baseline of $\mu = 72$ bpm. A sample of 36 participants after the program has mean resting heart rate $\bar{x} = 68.5$ bpm with sample standard deviation $s = 10.5$ bpm. State the hypotheses (choosing a two-sided alternative), compute the test statistic, and using a p-value of approximately 0.11, state a conclusion at $\alpha = 0.05$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

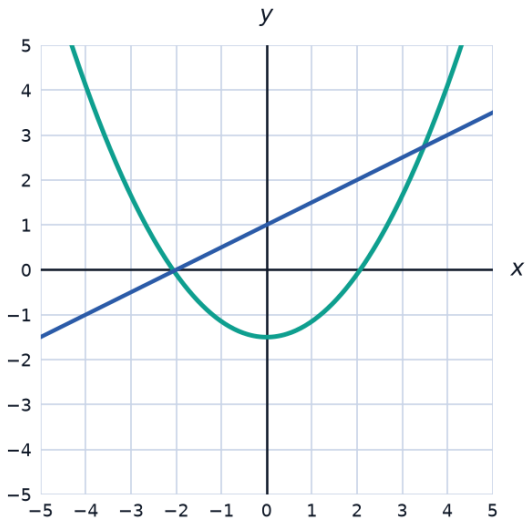
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15 [Challenging] • Target: One-sample t-test for a mean • Cross-Lesson: 2.01 Sampling Methods & Bias

Explain the relationship between a confidence interval and a two-sided hypothesis test at the same confidence/significance level — specifically, describe how you could use a 95% confidence interval alone to decide whether to reject $H_0 : \mu = \mu_0$ at $\alpha = 0.05$, without separately computing a p-value.

16 [Challenging] • Target: One-sample t-test for a mean • Cross-Lesson: 2.02 Experimental Design & Scope of Inference

A city council is deciding whether to fund a new public transit line, and wants to survey residents to estimate the true proportion who would use it regularly. A preliminary small survey suggests around 35% support, but the council wants a formal survey with a margin of error no larger than 2.5% at 95% confidence. Using $n = \left(\frac{z^*}{m}\right)^2 p(1-p)$ with the preliminary estimate $p = 0.35$ and $z^* = 1.96$, find the required sample size, rounding appropriately.



17 [Challenging] • Target: One-sample t-test for a mean • Cross-Lesson: 3.01 Probability Rules & Conditional Probability

A restaurant chain wants to test whether a new menu item's average preparation time is longer than the target of 8 minutes, since customers have complained about slow service. A random sample of 30 orders for the new item has a mean preparation time of $\bar{x} = 8.7$ minutes with sample standard deviation $s = 1.8$ minutes. State the hypotheses for this one-sided test, compute the test statistic, and using a p-value of approximately 0.019, state a conclusion at $\alpha = 0.05$ in the context of the restaurant's concern.

18 [Challenging] • Target: Margin of error and sample size • Cross-Lesson: 3.02 Discrete Random Variables & Binomial Distributions

A random variable X has distribution $\text{Po}(\lambda)$. A single observation of X is taken to test $H_0 : \lambda = 6$ against $H_1 : \lambda < 6$ at the 5% level. (a) Find the critical region. (b) State the actual significance level. (c) Find the probability of a Type II error if the true value of λ is 3.

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19 [Challenging] • Target: The logic of hypothesis testing • Cross-Lesson: 3.03 The Normal Distribution

Level 3 · AdvancedA school administrator randomly surveys 120 high school seniors at Oakwood High School. In the sample, 65% report planning to attend a four-year college, with a margin of error of $\pm 4\%$. Which of the following conclusions is most statistically sound? (A) Exactly 65% of all Oakwood High seniors will attend college. (B) It is plausible that between 61% and 69% of all Oakwood High seniors plan to attend a four-year college. (C) Between 61% and 69% of all high school seniors in the state plan to attend college. (D) To reduce the margin of error by half, the administrator should survey 240 seniors.

20 [Challenging] • Target: Four-step hypothesis testing procedure • Cross-Lesson: 3.04 Sampling Distributions & Central Limit Theorem

Level 4 · SynthesisA random sample of 400 registered voters in a city of 50,000 voters found that 55% support a transit bond, with an associated margin of error of 4.8% at the 95% confidence level. If a second survey uses a random sample of 1,600 registered voters from the same city, what will be the new margin of error?

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Constructing a confidence interval

A sample of $n = 121$ has mean $\bar{x} = 393$. A claimed population mean is $\mu_0 = 409$, with sample standard deviation $s = 62$. Compute the test statistic $z = (\bar{x} - \mu_0)/(s/\sqrt{n})$, to two decimal places.

a. 2.84

b. -2.34

c. -3.34

d. -2.84

22 [Foundational] • Target: Margin of error and sample size

A random sample of $n = 121$ has mean $\bar{x} = 180$ and sample standard deviation $s = 16$. Construct a 99% confidence interval for the population mean, using $z^* \approx 2.576$.

a. $(176.25, 181.75)$

b. $(175.25, 184.75)$

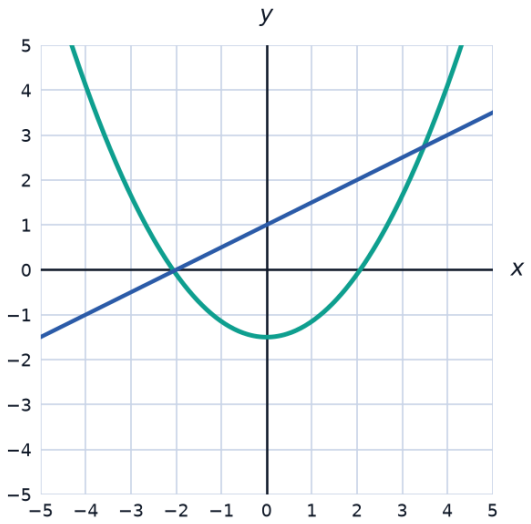
c. $(178.25, 183.75)$

d. $(176.25, 183.75)$

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23 [Foundational] • Target: The logic of hypothesis testing

A researcher wants a margin of error of at most 0.04 for a 99% confidence interval for a population proportion, using a planning value of $p = 0.30$. Find the minimum sample size needed, using $n = (z^*/m)^2 p(1-p)$ with $z^* = 2.576$.



a. 821

b. 870

c. 871

d. 921

24 [Foundational] • Target: Four-step hypothesis testing procedure

A hypothesis test about a manufacturer's claimed mean product lifetime produces a p-value of 0.047. Using a significance level of $\alpha = 0.01$, state the correct conclusion.

a. The test is inconclusive and must be repeated

b. Fail to reject H_0; not enough evidence that the true mean lifetime differs from the claim

c. Accept H_0 as true

d. Reject H_0; the true mean lifetime differs from the claim

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25 [Foundational] • Target: Type I and Type II errors, and power

A hypothesis test about a nutritionist's claimed mean calorie count produces a p-value of 0.112. Using a significance level of $\alpha = 0.01$, state the correct conclusion.

a. The test is inconclusive and must be repeated

b. Fail to reject H_0; not enough evidence that the true mean calorie count differs from the claim

c. Reject H_0; the true mean calorie count differs from the claim

d. Accept H_0 as true

26 [Foundational] • Target: Confidence intervals for a proportion

A hypothesis test about a claimed mean test score produces a p-value of 0.089. Using a significance level of $\alpha = 0.05$, state the correct conclusion.

a. Accept H_0 as true

b. Fail to reject H_0; not enough evidence that the true mean test score differs from the claim

c. Reject H_0; the true mean test score differs from the claim

d. The test is inconclusive and must be repeated

27 [Foundational] • Target: Determining sample size for a target margin of error

A hypothesis test about a claimed mean reaction time produces a p-value of 0.034. Using a significance level of $\alpha = 0.05$, state the correct conclusion.

a. The test is inconclusive and must be repeated

b. Accept H_0 as true

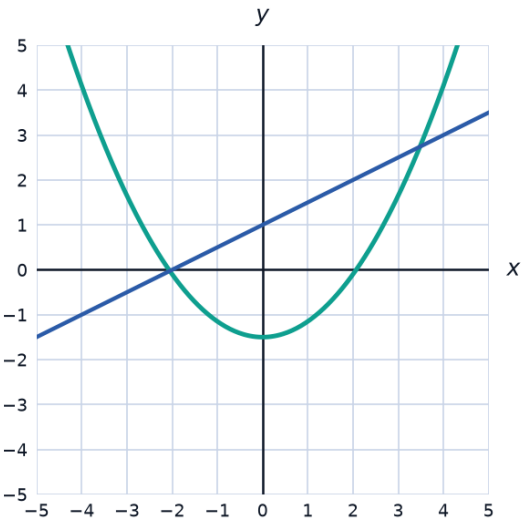
c. Fail to reject H_0; not enough evidence that the true mean reaction time differs from the claim

d. Reject H_0; the true mean reaction time differs from the claim

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28 [Foundational] • Target: One-sample t-test for a mean

A poll of $n = 100$ people finds $\hat{p} = 0.22$. Find the margin of error for a 90% confidence interval for the population proportion ($z^* = 1.645$).



a. 0.0681

b. 0.0381

c. 0.1181

d. 0.1381

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

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29 [Intermediate] • Target: Constructing a confidence interval

A quality control protocol uses a two-stage hypothesis test to monitor the rate of flaws, λ per metre, in the production of high-grade optical fibre. The number of flaws in a length of fibre follows a Poisson distribution. The manufacturer tests: $H_0 : \lambda = 1 \quad \text{against} \quad H_1 : \lambda > 1.$ The two-stage procedure operates as follows: - **Stage 1:** A $1-\text{metre}$ sample is inspected, and the number of flaws X_1 is recorded. If $X_1 \geq 3$, H_0 is rejected immediately. If $X_1 = 0$, H_0 is accepted immediately. If $X_1 \in \{1, 2\}$, the test proceeds to Stage 2. - **Stage 2:** A second, independent $1-\text{metre}$ sample is inspected, and the number of flaws X_2 is recorded. If $X_1 + X_2 \geq 4$, H_0 is rejected; otherwise, H_0 is accepted. Given that the true flaw rate is $\lambda = 2$ per metre, find the probability that a Type II error is committed.

a. $\mathrm{e}^{-2} + 16\mathrm{e}^{-4} \approx 0.428$

b. $16\mathrm{e}^{-4} \approx 0.293$

c. $4\mathrm{e}^{-2} \approx 0.541$

d. $1 - \mathrm{e}^{-1} - \frac{7}{2}\mathrm{e}^{-2} \approx 0.158$

30 [Intermediate] • Target: Margin of error and sample size

A manufacturer claims a machine produces parts with a mean thickness of 5.0 mm. A quality control inspector takes a random sample of 100 parts and finds a 95% confidence interval for the mean thickness of $(5.02, 5.08)$ mm. What does this indicate about the manufacturer's claim?

a. The claim is fully supported because 5.0 is close to the interval.

b. The claim is not supported at the 95% confidence level because 5.0 falls outside the interval.

c. The inspector must take another sample.

d. The machine definitely produces parts with mean 5.05 mm.

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31 [Intermediate] • Target: The logic of hypothesis testing

In estimating a population proportion, what is the effect on the margin of error of increasing the confidence level (with the same sample size), assuming all else is held constant?

a. The margin of error tends to increase

b. The margin of error tends to decrease

c. The margin of error stays exactly the same

d. The margin of error becomes zero

32 [Intermediate] • Target: Four-step hypothesis testing procedure

A sample of $n = 300$ gives $\hat{p} = 0.49$. Construct a 99% confidence interval for the population proportion ($z^* = 2.576$).

a. (0.3785, 0.6015)

b. (0.4528, 0.5272)

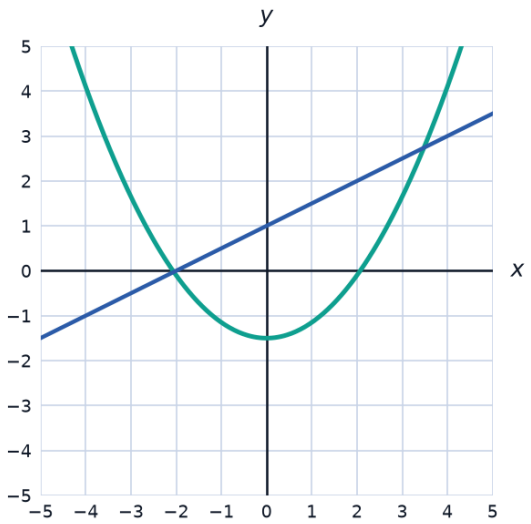
c. (0.4157, 0.5643)

d. (0.3413, 0.6387)

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33 [Intermediate] • Target: Type I and Type II errors, and power

A machine produces steel rods whose lengths are normally distributed with standard deviation $\sigma = 2.4$ cm. A quality controller suspects that the mean length has increased from the target value of 50 cm. A random sample of 36 rods is measured, yielding a sample mean of $\bar{x} = 50.8$ cm. Find the value of the standardized test statistic z for testing $H_0 : \mu = 50$ against $H_1 : \mu > 50$.



a. 2.0

b. 0.33

c. 1.2

d. 0.5

34 [Intermediate] • Target: Confidence intervals for a proportion

A researcher computes a 90% confidence interval and a 99% confidence interval from the same sample data. How do their widths compare?

a. The 90% interval is wider.

b. The 99% interval is wider.

c. Both intervals have the same width.

d. It depends on the sample mean.

35 [Intermediate] • Target: Determining sample size for a target margin of error

The mean of five test scores is 82. If four of the scores are 80, 85, 75, and 90, what is the fifth score?

a. 78

b. 80

c. 82

d. 84

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36 [Intermediate] • Target: One-sample t-test for a mean

A confidence interval for the proportion of defective items produced by a machine is $(0.02, 0.08)$. What is the point estimate of the proportion?

a. 0.03

b. 0.04

c. 0.05

d. 0.06

37 [Intermediate] • Target: Constructing a confidence interval

A 95% confidence interval for a population proportion is $(35\%, 41\%)$. What is the margin of error?

a. 2\%

b. 11\%

c. 3\%

d. 8\%

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Margin of error and sample size • Cross-Lesson: 3.02 Discrete Random Variables & Binomial Distributions

A 95% confidence interval for a population proportion is $(58\%, 64\%)$. What is the point estimate?

a. 59\%

b. 61\%

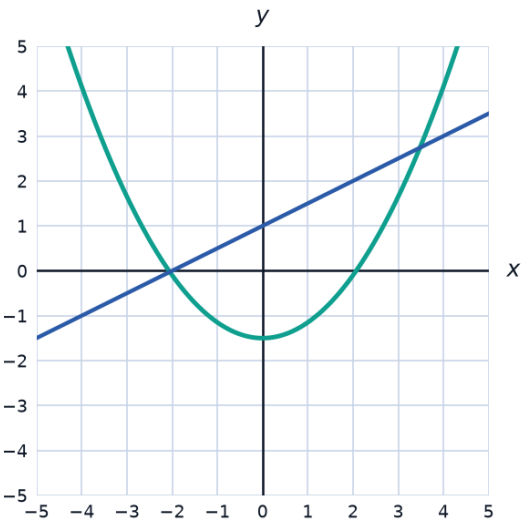
c. 71\%

d. 63\%

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39 [Challenging] • Target: The logic of hypothesis testing • Cross-Lesson: 3.03 The Normal Distribution

A hypothesis test yields a p-value of 0.173. What does this p-value mean?



a. H_0 should be rejected 17.3\% of the time

b. If H_0 is true, there is a 17.3\% chance of observing a test statistic at least this extreme

c. There is a 17.3\% chance that H_0 is true

d. The probability that H_a is true is 17.3\%

40 [Challenging] • Target: Four-step hypothesis testing procedure • Cross-Lesson: 3.04 Sampling Distributions & Central Limit Theorem

A sample of $n = 18$ has $\bar{x} = 76$, $s = 15$. Construct a 99% confidence interval for μ ($t^* = 2.898$).

a. (55.51, 96.49)

b. (65.75, 86.25)

c. (60.63, 91.37)

d. (70.88, 81.12)

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41 [Challenging] • Target: Type I and Type II errors, and power • Cross-Lesson: 4.02 Chi-Square Tests & Regression Inference

A survey was conducted on a sample of 500 randomly selected adults regarding their preferred mode of transportation for commuting to work. The results are summarized in the table below: | Mode of Transportation | Frequency | |---|---| | Car | 210 | | Public Transit | 150 | | Bicycle | 90 | | Walking | 50 | If the margin of error for this survey is $\pm 4\%$, and assuming the sample is representative of the population, which of the following statements is most accurate about the true proportion of adults in the population who prefer public transit? A) The true proportion is between 26\% and 34\%. B) The true proportion is between 22\% and 30\%. C) The true proportion is between 30\% and 38\%. D) The true proportion is exactly 30\%.

a. A

b. B

c. C

d. D

42 [Challenging] • Target: Confidence intervals for a proportion • Cross-Lesson: 1.01 Measures of Center & Spread

A machine produces metal rods whose lengths are normally distributed with unknown mean μ cm and known standard deviation $\sigma = 0.8$ cm. A quality control manager tests the hypotheses $H_0 : \mu = 25.0 \quad \text{against} \quad H_1 : \mu < 25.0$ at the 5% level of significance. A random sample of 16 rods yields a sample mean length of $\bar{x} = 24.6$ cm. Given that the critical value for this test is $z_{\text{crit}} = -1.645$, which of the following gives the correct test statistic z and the corresponding conclusion?

a. $z = -2.0$; reject H_0 because $-2.0 < -1.645$

b. $z = -2.0$; do not reject H_0 because $-2.0 < -1.645$

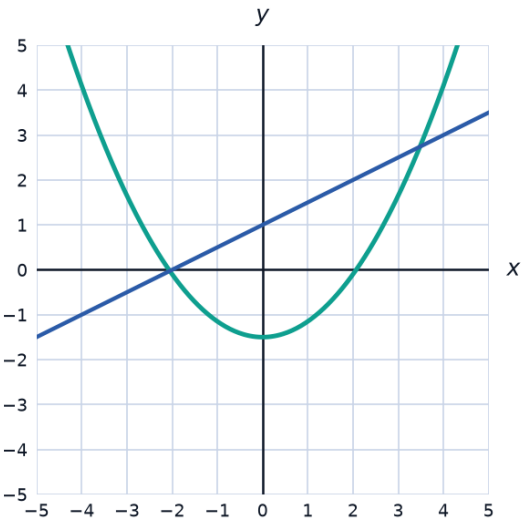
c. $z = -0.5$; do not reject H_0 because $-0.5 > -1.645$

d. $z = -8.0$; reject H_0 because $-8.0 < -1.645$

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43 [Challenging] • Target: Determining sample size for a target margin of error • Cross-Lesson: 1.02 Z-Scores & Linear Transformations

The mean of 5 numbers is 12. Four of them are 8, 10, 14, 16. What is the fifth?



a. 10

b. 12

c. 14

d. 16

44 [Challenging] • Target: One-sample t-test for a mean • Cross-Lesson: 1.03 Describing Distributions & Linear Regression

A class of 20 students had a mean score of 80 on a test. One student's score of 95 was mistakenly recorded as 55. When the error is corrected, what is the new mean score?

a. 81

b. 82

c. 83

d. 84

45 [Challenging] • Target: Constructing a confidence interval • Cross-Lesson: 2.01 Sampling Methods & Bias

A confidence interval for the mean uses $z = 1$, $\sigma = 4$, $n = 9$. Find the margin of error $E = z\sigma/\sqrt{n}$.

a. $\frac{7}{3}$

b. $\frac{4}{3}$

c. $\frac{17}{12}$

d. $\frac{8}{3}$

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